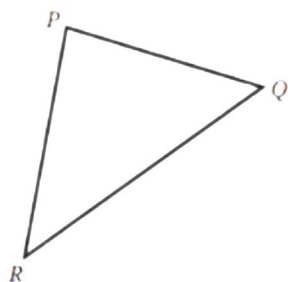


1



Note: Figure not drawn to scale.

In triangle PQR , the measure of $\angle P$ is 47° , and the measure of $\angle Q$ is 83° . What is the measure of $\angle R$?

(A) 36°

(B) 50°

(C) 94°

(D) 130°

2

	Live north of Center St.	Live south of Center St.	Total
Less than 45 years old	16	13	29
At least 45 years old	19	87	106
Total	35	100	135

The table summarizes members of a local organization by age and whether they live north or south of Center St. If a member of the organization is selected at random, what is the probability that the selected member is at least 45 years old?

(A) $\frac{29}{135}$

(B) $\frac{35}{135}$

(C) $\frac{100}{135}$

(D) $\frac{106}{135}$

3

If $3x + 4 = 10$, what is the value of $9x - 4$?

4

The function f is defined by $f(x) = 2x^2 - 5x$. What is the value of $f(8)$?

5

Line q in the xy -plane has a slope of -9 and passes through the point $(0, 14)$. Which equation defines line q ?

(A) $y = -9x + 14$

(B) $y = 9x + 14$

(C) $y = 14x - 9$

(D) $y = 14x + 9$

6

For a 2-week period in a town in Iowa, the lowest recorded temperature was 26 degrees Fahrenheit ($^{\circ}\text{F}$) and the highest recorded temperature was 63°F . Which inequality is true for any recorded temperature t , in $^{\circ}\text{F}$, in this town for this 2-week period?

(A) $t \leq 26$

(B) $26 \leq t \leq 63$

(C) $t \geq 63$

(D) $t \geq 89$

7

$$f(x) = -4x^2 + bx + c$$

In the given quadratic function, b and c are constants. The graph of $y = f(x)$ is a parabola that has a vertex at the point (h, k) , where h and k are constants. If $k < 0$ and $f(-2) = f(6)$, which of the following must be true?

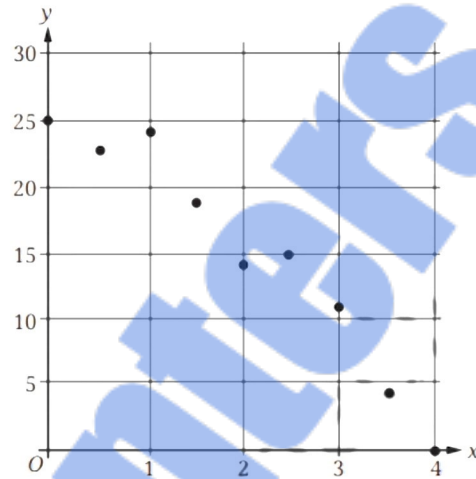
- I. $c < -15$
- II. $b \geq 15$
- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

8

A student council group is selling car stickers for a fundraiser. They use the function $p(x) = 5x - 190$ to determine their profit $p(x)$, in dollars, for selling x car stickers. In order to earn a profit of \$600, how many car stickers must they sell?

9

The scatterplot shows the relationship between two variables, x and y .



Which of the following equations is the most appropriate linear model for the data shown?

☐ (A) $y = -6 + 27x$

☐ (B) $y = 6 - 27x$

☒ (C) $y = 27 + 6x$

☐ (D) $y = 27 - 6x$

10

A batch of smoothies consists of 4 cups of milk and 2 bananas and has 1,226 milligrams(mg) of calcium. There is 304 mg of calcium in 1 cup of milk. How much calcium, in mg, is in 1 banana?

(A) 5

(B) 10

(C) 461

(D) 1,216

11

$$f(x) = 7,000(0.90)^x$$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 7,000 in this context?

(A) The estimated initial insect population for this species and area in 2008

(B) The estimated insect population for this species and area 8 years after 2008

(C) The estimated percent decrease in the insect population for this species and area each year after 2008

(D) The estimated percent decrease in the insect population for this species and area every 8 years after 2008

12

$$7rx^2 - 28sx + 7r$$

In the given equation, r is a positive constant, and s is a negative constant. The equation has two solutions. If $\frac{r}{s} > k$, what's the minimum value of k ?

13

The measure of angle F is $\frac{7\pi}{4}$ radians. What is the value of $\cos(F)$?

(A) $-\frac{\sqrt{2}}{2}$

(B) $-\frac{1}{2}$

(C) $\frac{\sqrt{2}}{2}$

(D) $\frac{\sqrt{3}}{2}$

14

Line k has a slope of 6 and a y -intercept of $(0, 60)$. What is the x -coordinate of the x -intercept of line k ?

15

A hemisphere is half of a sphere. If a hemisphere has a radius of 56 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- (A) 6,600
- (B) 26,300
- (C) 275,900
- (D) 367,800

16

The function g is defined by $g(x) = -\frac{x}{3} + 11$. What is the x -intercept of the graph of $y = g(x) + 6$ in the xy -plane?

- (A) $(-33, 0)$
- (B) $(-51, 0)$
- (C) $(33, 0)$
- (D) $(51, 0)$

17

An auditorium has seats for 2,600 people. Tickets to attend a show at the auditorium currently cost \$8.00. For each \$1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,800x + 20,800$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

- (A) 8
- (B) 9
- (C) 18
- (D) 26

18

An event planner is planning an ice-skating event. It costs the event planner a onetime fee of \$30 to rent the skating rink and \$10.25 per attendee. The event planner has a budget of \$250. What is the greatest number of attendees possible without exceeding the budget?

19

The function f gives the product of a number, x , and a number that is 45 more than x . Which equation defines f ?

(A) $f(x) = x^2 + x + 45$

(B) $f(x) = x^2 + 45$

(C) $f(x) = x^2 + 45x$

(D) $f(x) = x^2 + 45x + 45$

22

An object's speed is increasing at a rate of 10.5 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

(A) 0.4

(B) 23.5

(C) 153.2

(D) 281.6

20

$$ax + by = 96$$

$$2ax + 8y = 72$$

In the given system of equations, a and b are constants. The graphs of these equations in the xy -plane intersect at the point $(x, 6)$. What is the value of b ?

(A) 4

(B) 6

(C) 8

(D) 14

21

If the perimeter of an equilateral triangle with a height of 21 is $a\sqrt{3}$, what is the value of a ?

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Module-2

1

The table summarizes the UV index value recorded by a research assistant at noon each day for 48 days.

UV index	Number of days
1	8
2	13
3	12
4	15

According to the table, a UV index value of 1 was recorded on how many days?

(A) 48

(B) 40

(C) 8

(D) 4

2

The function j is defined by $j(x) = \frac{x}{7}$. What is the value of $j(42)$?

(A) 42

(B) 35

(C) 7

(D) 6

3

The speed of a white-throated needletail, a type of bird, in flight was measured to be 79 miles per hour. What was the white-throated needletail's measured speed, in kilometers per hour? (Use 1 mile = 1.6 kilometers.)

(A) 126.4

(B) 80.6

(C) 77.4

(D) 49.4

4

A company developed a plan to set the selling price of a product. The company determined that for a selling price of \$150.00, zero products would be sold. Each \$2.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly \$1,687.50, which of the following could be the number of products sold?
(revenue = price $\times$ number of products sold)

5

What is the y-intercept of the graph of $y = 11^x$ in the xy-plane?

☐ (A) (0, 1)

☐ (B) (0, 0)

☐ (C) (0, 11)

☐ (D) (0, 10)

6

$$y = x + 19$$

$$y = -x + 33$$

The solution to the given system of equations is (x, y) . What is the value of $2y$?

☐ (A) 14

☐ (B) 26

☐ (C) 52

☐ (D) 104

7

To cut a lawn, Antwan charges a fee of \$15.00 for his equipment and \$8.50 per hour spent cutting a lawn. Taylor charges a fee of \$12.00 for her equipment and \$9.00 per hour spent cutting a lawn. If x represents the number of hours spent cutting a lawn, what are all the values of x for which Taylor's total charge is greater than Antwan's total charge?

☐ (A) $5 \leq x \leq 6$
☐ (B) $6 \leq x \leq 7$
☐ (C) $x < 5$
☐ (D) $x > 6$

8

The table summarizes the number of objects in each group.

Group	Number of objects
A	375
B	54
C	690
D	81
Total	1,200

The number of objects in group C is $p\%$ of the number of objects in group A. What is the value of p ?

9

Which quadratic equation has exactly one distinct real solution?

(A) $-97x^2 = 0$

(B) $9x^2 + 3 = 0$

(C) $9x^2 + 3x + 1 = 0$

(D) $-97x^2 - 97x = 0$

10

$$x^2 + 9x + 5 = 0$$

One solution to the given equation can be written as

$$x = \frac{-9 + \sqrt{k}}{2}, \text{ where } k \text{ is a constant. What is the value}$$

of k ?

12

The function f is defined by $f(x) = 9x^2 + kx - 21$, where k is a constant. In the xy -plane, the graph of $y = f(x)$ passes through the points $(r, 23)$ and $(2r, 139)$, where r is a constant. Which of the following could be the value of k ?

(A) 4

(B) 12

(C) 116

(D) 162

11

The table gives the distribution of favorite core subject and grade level for 80 students.

Subject	Grade level			
	Sixth	Seventh	Eighth	Total
English	4	9	9	22
Mathematics	10	1	9	20
Science	4	6	4	14
Social studies	6	9	9	24
Total	24	25	31	80

If a student is selected at random, what is the probability of selecting a student whose favorite core subject is mathematics?

(A) $\frac{1}{10}$

(B) $\frac{1}{5}$

(C) $\frac{1}{4}$

(D) $\frac{2}{3}$

14

The function f is defined by $f(x) = 41(0.28)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p ?

(A) 72

(B) 59

(C) 41

(D) 28

13

A rectangular poster has an area of 360 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 40%. What is the area of the copy, in square inches?

16

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(35) > 0$, $f(38) < 0$, and $f(41) > 0$, which of the following could be the value of $a + b$?

(A) 38

(B) 40

(C) 73

(D) 75

17

$$\sqrt{x^2 - 20x + 100} - 11x = 0$$

What are all possible solutions to the given equation?

(A) -1

(B) $\frac{5}{6}$

(C) -1 and 1

(D) -1 and $\frac{5}{6}$

15

A town manager selected 2 random samples of residents in the town to estimate the percentage of all residents that planned to vote in favor of a new business in the downtown area. Based on the first sample, he estimated that 75% of the residents in the town planned to vote in favor, with an associated margin of error of 8.5%. Based on the second sample, he estimated that 83% of the residents in the town planned to vote in favor, with an associated margin of error of 6.9%. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained a larger margin of error than the second sample?

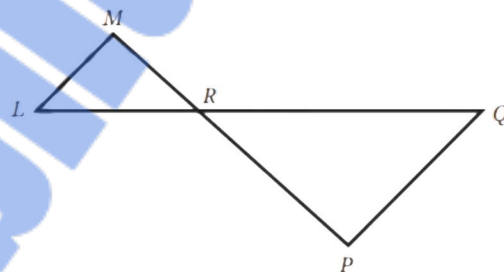
(A) The first sample contained more residents than the second sample.

(B) The first sample contained fewer residents than the second sample.

(C) The first sample contained a higher percentage of residents that planned to vote in favor of a new business in the downtown area.

(D) The first sample contained a lower percentage of residents that planned to vote in favor of a new business in the downtown area.

18



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 9 and 17 units, respectively. The area of $\triangle LMR$ is 45 square units. What is the area of $\triangle PQR$, in square units?

(A) $\frac{3.645}{289}$ (B) $\frac{405}{17}$

(C) 85

(D) $\frac{1,445}{9}$

19

$$6x^2(3x - 16)(3x - u) = 0$$

In the given equation, u is a positive constant. The sum of the solution to the equation is 13. What is the value of u ?

20

The cost of renting a power washer is \$44 for the first day and \$22 for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the power washer for d days, where d is a positive integer?

(A) $C(d) = 22d + 22$

(B) $C(d) = 22d + 44$

(C) $C(d) = 44d - 22$

(D) $C(d) = 44d + 66$

21

Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(9, -7)$, contains the point $(8, -9)$, and has a y -intercept at $(0, a)$. The graph of $y = 6 \cdot f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

22

A 9-pound cat eats two types of canned cat food: chicken-flavored and duck-flavored. The recommended amount of chicken-flavored food is 2.25 cans per 8 pounds of a cat's weight per day. The recommended amount of duck-flavored food is 0.75 cans per 5 pounds of a cat's weight per day. If c is the number of chicken-flavored cans and d is the number of duck-flavored cans a 9-pound cat eats in one day, which equation describes all the possible values of c and d for the cat to eat the recommended amount of food based on its weight?

(A) $\frac{0.75}{5}c + \frac{2.25}{8}d = 9$

(B) $\frac{2.25}{8}c + \frac{0.75}{5}d = 9$

(C) $\frac{5}{0.75}c + \frac{8}{2.25}d = 9$

(D) $\frac{8}{2.25}c + \frac{5}{0.75}d = 9$